

Erectile Dysfunction

Updated with Recent Urology Guidelines

Definition & Epidemiology

- Erectile dysfunction (ED): persistent inability to achieve or maintain an erection sufficient for satisfactory sexual performance.
 - Affects ~30 million men in the United States; prevalence increases with age.
 - Often associated with cardiovascular, metabolic, neurologic, or psychological conditions.
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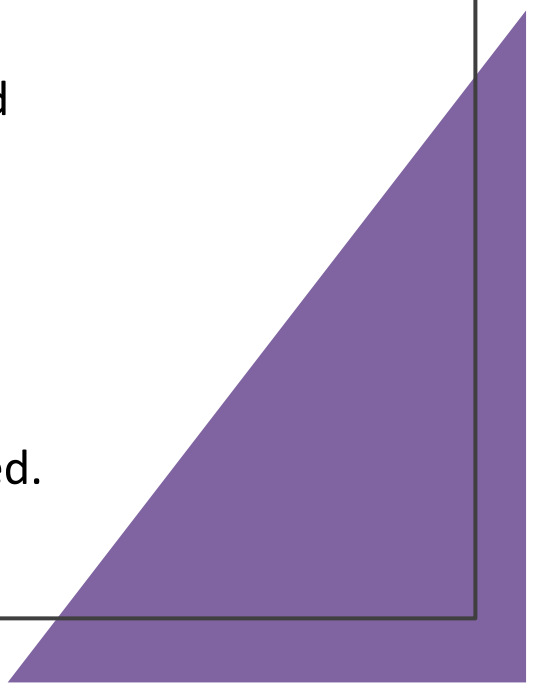
Etiology of Erectile Dysfunction

- Vascular: atherosclerosis, hypertension, diabetes.
- Neurogenic: spinal cord injury, neuropathy.
- Hormonal: hypogonadism, thyroid disorders.
- Medication-induced: antihypertensives, SSRIs.
- Psychogenic factors: anxiety, depression.

Pathophysiology

- Erection depends on nitric oxide–mediated smooth muscle relaxation.
- Endothelial dysfunction impairs vasodilation.
- Diabetes and cardiovascular disease reduce penile blood flow.

Diagnostic Evaluation

- Thorough medical and sexual history.
 - Physical examination focused on cardiovascular and neurological systems.
 - Laboratory tests: fasting glucose, lipid profile, testosterone level.
 - Nocturnal penile tumescence testing when indicated.
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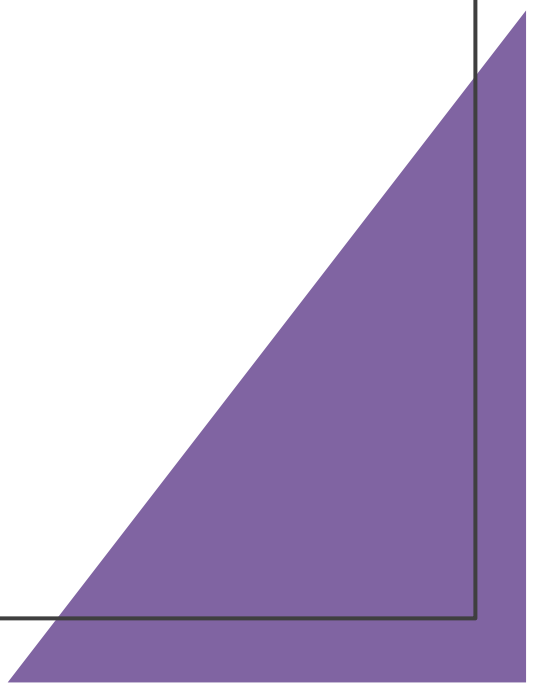
Management Overview

- Lifestyle modification: weight loss, smoking cessation, exercise.
- Optimize management of comorbid diseases (diabetes, hypertension).
- Pharmacologic therapy is first-line in most cases.

Pharmacologic
Treatment (Based on
Pharmacotherapy
Handbook)

- PDE5 inhibitors: sildenafil, tadalafil, vardenafil.
- Onset and duration vary by agent; tadalafil longest duration.
- Contraindicated with nitrates due to severe hypotension risk.
- Adverse effects: headache, flushing, dyspepsia, nasal congestion.

Updated Urology Recommendations (AUA/EAU 2023–2024)

- First-line therapy: PDE5 inhibitors unless contraindicated.
 - Assess cardiovascular risk before initiating ED treatment.
 - Testosterone therapy only for confirmed hypogonadism.
 - Second-line options: intracavernosal injections (alprostadil), vacuum erection devices.
 - Penile prosthesis recommended for refractory cases.
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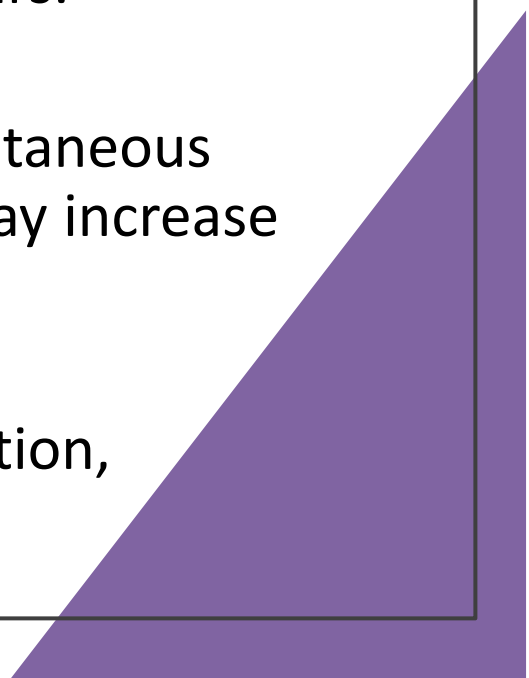
1. Phosphodiesterase Type 5 Inhibitors

- ✓ Act by selectively inhibiting phosphodiesterase (PDE) type 5, an enzyme that breaks down cGMP.
- ✓ By inhibiting the breakdown of cGMP, smooth muscle relaxation is induced, leading to an erection.
- ✓ However, the PDE inhibitors are only effective in the presence of sexual stimulation to drive the nitric oxide/cGMP system, making them facilitators of an erection, not initiators.

1. Phosphodiesterase Type 5 Inhibitors...

- Pts with radical prostatectomy , DM, severe nerve damage, or severe vascular disease have lower response rates.
 - Considered first line therapies due to high efficacy rates, convenience of dosing & minimal severe adverse effects.
 - The most dramatic difference b/n the three agents is tadalafil's extended duration of action, earning it the nickname "the weekender drug."
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1. Phosphodiesterase Type 5 Inhibitors...

- Sildenafil & vardenafil have average half-lives of 3 to 4 hrs, tadalafil's half life is approximately 18 hrs.
 - The extended half-life allows for more spontaneous sexual activity over a couple of days, but may increase the duration of ADE & likelihood of DIs.
 - S/E: headache, facial flushing, nasal congestion, dyspepsia & rarely priapism.
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1. Phosphodiesterase Type 5 Inhibitors...

- Vardenafil & sildenafil may also cause difficulty in discriminating blue from green, bluish tones in vision, or difficulty seeing in dim light due to cross-reactivity with PDE 6 in the retina.
- Non arteritic ischemic optic neuropathy (NAION) in a small number of pts, a condition in which blood flow is blocked to the optic nerve.
- PDE inhibitors can lead to significant hypotension.
- Pts taking organic nitrates are the most at risk, as they potentiate the drop in blood pressure.

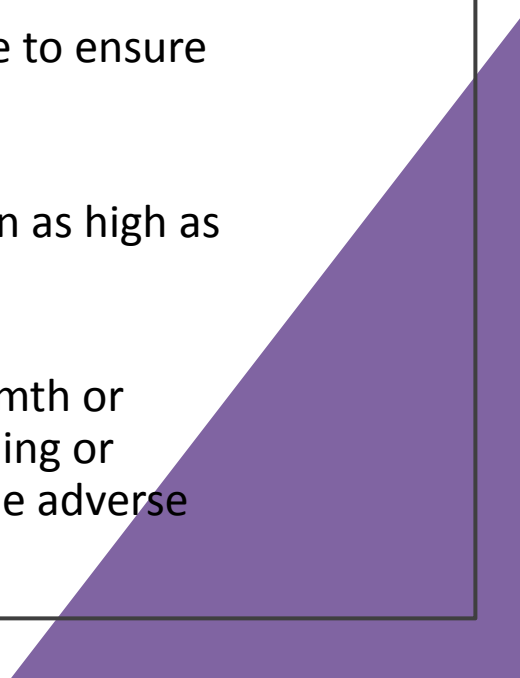
1. phosphodiesterase Type 5 Inhibitors...

- Caution should be used when using a PDE inhibitor in pts taking α -blockers due to an increased risk of hypotension.
- Vardenafil may possibly cause QT prolongation.
- 6 to 8 attempts with a medication & specific dose may be needed before successful intercourse results.

2. Alprostadil

- Alprostadil is a PGE1 analog that induces an erection by stimulating adenyl cyclase, which leads to an increase in smooth muscle relaxation, rapid arterial inflow, and increased penile rigidity.
- Alprostadil is available as an intra cavernosal injection (Caverject® or Edex®) or a transurethral suppository (MUSE®, medicated urethral system for erection), but the injectable form is more effective.
- Both forms of alprostadil are considered more invasive than oral medications or vacuum Erection Devices (VEDs), and are therefore second-line therapies.

2. Alprostadil...

- MUSE consists of a urethral pellet of alprostadil with an applicator.
 - Onset of action is within 5 to 10 min & it is effective for 30 to 60 min.
 - Initial dose titration should occur in a physician's office to ensure correct dose & prevent adverse events.
 - Although effectiveness rates in clinical trials have been as high as 65%, its success in practice has been lower.
 - Aching in the penis, testicles, legs, and perineum, warmth or burning sensation in the urethra, minor urethral bleeding or spotting, priapism, and lightheadedness are all possible adverse effects.
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2.Alprostadil

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- Partners may experience vaginal burning or itching.
- Disadvantages : lower effectiveness, high cost, adverse effects, complicated insertion technique & a contraindication against use with a pregnant partner unless using a condom.
- Alprostadil injected into the corpus cavernosum is the more effective form and is the only FDA-approved agent for injection.
- The onset of action is similar to transurethral alprostadil, but duration varies with dose & must be titrated in a physician's office to achieve an erection lasting no more than 1 hr

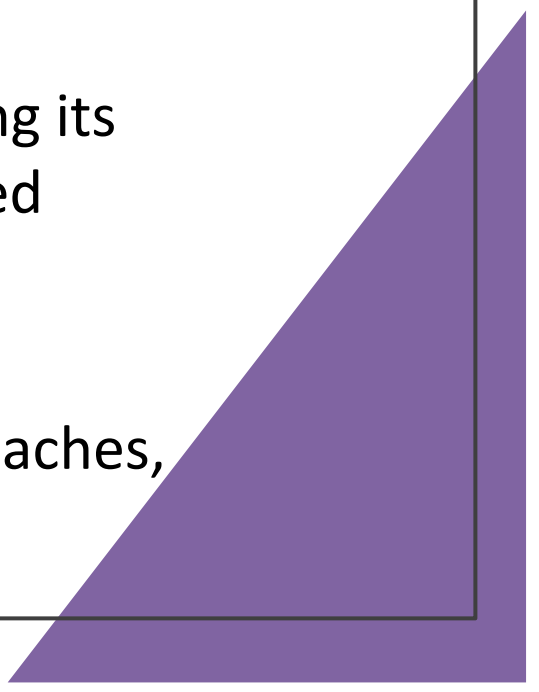
2. Alprostadil...

- Injections should be done into one side of the penis directly into the corpus cavernosum, and then the penis should be massaged to distribute the drug.
- Intracavernosal injections are effective in up to 90% of pts.
- Adverse effects: pain with injection, bleeding or bruising at the injection site, fibrosis, or priapism.
- Use with caution in pts with sickle cell disease, those on anticoagulants, or those who have bleeding disorders, due to an increased risk of priapism and bleeding.

3. Papaverine and Phentolamine

- Papaverine is a non-selective PDE inhibitor that induces an erection by relaxing smooth muscle & increasing blood flow.
- Phentolamine is a competitive α -adrenergic receptor antagonist that increases arterial inflow by opposing arterial constriction.
- Both drugs are rarely used alone, and are most often mixed in various conc with alprostadil for increased effectiveness & to reduce adverse effects with smaller doses of each medication.

4. Yohimbine

- It selectively inhibits α 2-adrenergic receptors in the brain that are associated with libido & penile erection.
 - Since there is only limited data supporting its efficacy, yohimbine is not a recommended treatment for any form of ED.
 - Adverse effects: nausea, irritability, headaches, anxiety, tachycardia, and hypertension.
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5. Testosterone Supplementation

- Only effective in pts with low serum testosterone .
- Testosterone replacement is the initial treatment of choice in hypogonadism, as it corrects decreased libido, fatigue, muscle loss, sleep disturbances, and depressed mood.
- Improvements in ED may occur, but they should not be expected to occur in all pts.
- The initial trial should be for 3 months.

5. Testosterone Supplementation...

- General side effects of testosterone :
 - gynecomastia, dyslipidemia, polycythemia & acne
 - Weight gain, hypertension, edema & exacerbations of CHF occur due to sodium retention
 - Before initiating testosterone, the pt should undergo evaluation for BPH & prostate Ca.
 - Routine follow up: yearly PSA, DRE, Hgb & LFT in addition to assessment of response.

Special Considerations

- Diabetes: reduced response to PDE5 inhibitors; may require higher doses.
 - Post-prostatectomy ED: early rehabilitation improves outcomes.
 - Psychogenic ED: combine pharmacologic therapy with counseling.
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OUTCOME EVALUATION

- Successful therapy for ED results in
 - an increase in erections suitable for intercourse.
 - Improvement in the pt's QoL.
- Laboratory evaluation & PE are not necessary for evaluation of effectiveness, but may be necessary to determine adverse events.
- Satisfaction & effectiveness are evaluated after a 4-wk trial unless the pt initiates follow-up sooner.
- Intracavernosal injections will require multiple visits over the long term to detect AEs.

References

- Pharmacotherapy Handbook, 7th Edition.
 - American Urological Association (AUA) Guidelines on Erectile Dysfunction, 2023.
 - European Association of Urology (EAU) Erectile Dysfunction Guidelines, 2024.
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